**1.** Given the line $\vec{r} = [12,-8,-4] + t[-3,4,2]$, find the intersection(s) with the xy plane.

**xy plane equation : z=0**

$$\mathbf{x = 12 - 3t}$$
$$\mathbf{y = -8 + 4t}$$
$$\mathbf{z = -4 + 2t \xrightarrow{z=0} t = 2}$$
$$\mathbf{P.O.I : (6,0,0)}$$

**2.** Find vector and parametric equations of the plane that contains the two intersecting lines ,

$\vec{r} = [3,-1,2] + s[4,0,1]$ and , $\vec{r} = [3,-1,2] + t[4,0,2]$.

$$\vec{\mathbf{r}} = [3,-1,2] + p[4,0,1] + q[2,0,1] \ ; p,q \in \mathbf{R}$$
$$\mathbf{x = 3 + 4p + 2q}$$
$$\mathbf{y = -1}$$
$$\mathbf{z = 2 + p + q}$$

**3.** Find a scalar equation of the plane that contains the origin and the point (2, -3, 2) and is perpendicular to the plane $x + 2y - z + 3 = 0$.

$$\vec{\mathbf{n}}_1 = [1,2,-1]$$
$$\overrightarrow{\mathbf{OP}} = [2,-3,2]$$
$$\vec{\mathbf{n}}_2 = \vec{\mathbf{n}}_1 \times \overrightarrow{\mathbf{OP}}$$
$$= [1,2,-1] \times [2,-3,2]$$
$$= [1,-4,-7]$$
$$\textbf{equation of plane : } \mathbf{x - 4y - 7z = 0}$$

**4.** Determine the parametric equations for the plane containing the 2 parallel lines:
$L_1 : \vec{r} = [0,1,3] + t[-6,-3,6]$ and $L_2 : \vec{r} = [-4,5,-4] + s[4,2,-4]$.

$$\vec{\mathbf{u}} = \overrightarrow{\mathbf{PQ}} = [0,1,3] - [-4,5,-4] = [4,-4,7]$$
$$\vec{\mathbf{m}}_1 = -3[2,1,-2] \ , \ \vec{\mathbf{m}}_2 = 2[2,1,-2]$$
$$\vec{\mathbf{v}} = [2,1,-2]$$
$$\textbf{equation of plane : } \mathbf{x = 4s + 2t}$$
$$\mathbf{y = 1 - 4s + t}$$
$$\mathbf{z = 3 + 7s - 2t}$$

**5.** Determine the **scalar** equation of the plane parallel to the plane $\pi_1 : 3x + y - 2z - 4 = 0$ and containing the point of intersection of lines $L_1 : x = 7 + 2s, y = 2 + s, z = -6 - 3s$ and $L_2 : \vec{r} = [3,9,13] + t[1,5,5]$.

$$L_1 : x = 7 + 2s \qquad L_2 : x = 3 + t$$
$$y = 2 + s \qquad y = 9 + 5t$$
$$z = -6 - 3s \qquad z = 13 + 5t$$
$$7 + 2s = 3 + t \longrightarrow 2s - t = -4 \quad (1)$$
$$2 + s = 9 + 5t \longrightarrow s - 5t = 7 \quad (2)$$
$$-6 - 3s = 13 + 5t \longrightarrow 3s + 5t = -19 \quad (3)$$
From (2) & (3) : $s = -3$ , $t = -2$

Check: $2s - t = -4$

$$2(-3) - (-2) = 4$$

$\boxed{\text{P.O.I is } (1,-1,3)}$

$$\pi : 3x + y - 2z + D = 0 \leftarrow \text{point}(1,-1,3)$$
$$3 - 1 - 6 + D = 0$$
$$D = 4$$
$$\boxed{\therefore \pi : 3x + y - 2z + 4 = 0}$$

**6.** Find the scalar equation of the plane that is perpendicular to the plane $\vec{r} = [4,-5,2] + s[2,1,3] + t[-1,4,0]$ and intersects it at the line $\vec{r} = [4,-5,2] + t[1,-1,1]$.

$$\vec{n_1} = [2,1,3] \times [-1,4,0] = [-12,-3,9] = -3[4,1,-3]$$
$$\vec{m} = [1,-1,1]$$
$$\vec{n} = \vec{n_1} \times \vec{m}$$
$$= [4,1,-3] \times [1,-1,1]$$
$$= [-2,-7,-5]$$
$$= -[2,7,5]$$

$$\pi : 2x + 7y + 5z + D = 0 \leftarrow (4,-5,2)$$
$$8 - 35 + 10 + D = 0$$
$$D = 17$$
$$\boxed{\therefore \pi : 2x + 7y + 5z + 17 = 0}$$

**7.** Consider the lines $l_1 : \vec{r} = [1,-2,4] + s[1,1,-3]$ and $l_2 : \vec{r} = [4,-2,k] + t[2,3,1]$ Determine the equation of the plane that contains $l_1$ and is parallel to $l_2$ .

$$\vec{n} = [1,1,-3] \times [2,3,1] = [10,-7,1]$$
$$10x - 7y + z + D = 0 \leftarrow (1,-2,4)$$
$$10 + 14 + 4 + D = 0$$
$$D = 28$$
$$\boxed{\therefore \pi : 10x - 7y + z + 28 = 0}$$

**8.** Find the equation of the plane that contains the line $\vec{r}=[3,1,0]+t[2,1,4]$ and is perpendicular to the plane $\pi:\vec{p}=[1,1,1]+k[1,0,5]+s[-4,2,3]$.

$$\vec{n_1}=[1,0,5]\times[-4,2,3]=[-10,-23,2]$$
$$\vec{m}=[2,1,4]$$
$$\vec{n}=\vec{n_1}\times\vec{m}$$
$$=[-10,-23,2]\times[2,1,4]$$
$$=[-94,44,36]$$
$$=2[-47,22,18]$$

$$\pi:-47x+22y+18z+D=0 \leftarrow (3,1,0)$$
$$-141+22+D=0$$
$$D=-119$$
$$\boxed{\therefore \pi:-47x+22y+18z-119=0}$$

**9.** Determine the measure of the obtuse angle to the nearest degree between the lines:
$l_1:\vec{r}=[1,-2,3]+t[4,1,-1]$ and $l_2:\ x=3-t\ ,\ y=t\ ,z=3+2t$.

$$\cos\theta=\frac{\vec{m_1}\bullet\vec{m_2}}{|\vec{m_1}||\vec{m_2}|}$$
$$=\frac{[4,1,-1]\bullet[-1,1,2]}{\left(\sqrt{16+1+1}\right)\left(\sqrt{1+1+4}\right)}$$
$$\cos\theta=\frac{-5}{\sqrt{108}}$$
$$\boxed{\theta\doteq 118.7^\circ}$$

**10.** Determine the Cartesian equation of the plane that contains the point **A**(2,-1,1) and is perpendicular to the line joining points **B**(-1,3,2) and **C**(4,0,-2).

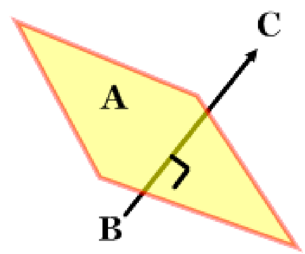
$\vec{n} = \overrightarrow{BC} = [5,-3,-4]$

$\pi : 5x - 3y - 4z + D = 0$

$A \in \pi : 10 + 3 - 4 + D = 0$

$D = -9$

$\boxed{\pi : 5x - 3y - 4z - 9 = 0}$




**11.** Determine the value(s) of k if the acute angle between the planes 2x+ky-8=0 and x-3y+z+4=0 is $60^\circ$.

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$\frac{1}{2} = \frac{[2,k,0]\cdot[1,-3,1]}{\left(\sqrt{4+k^2}\right)\left(\sqrt{1+9+1}\right)}$$

$$\frac{1}{2} = \frac{2-3k}{\sqrt{11(4+k^2)}} \qquad (2-3k>0 \text{ or } k<\frac{2}{3})$$

$$11(4+k^2) = 4(2-3k)^2$$

$$44 + 11k^2 = 16 - 48k + 36k^2$$

$$0 = 25k^2 - 48k - 28$$

$$k = \frac{24 \pm 2\sqrt{319}}{25}$$

$$k \doteq 2.39 \text{ or } \boxed{k \doteq -0.47}$$